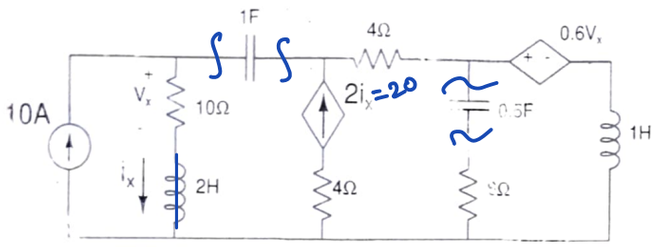
Department of Electrical Engineering - IIT Madras
EE2015 - Electric Circuits and Networks - Quiz 1
8 - 8:50 pm. August 28, 2024 20 marks

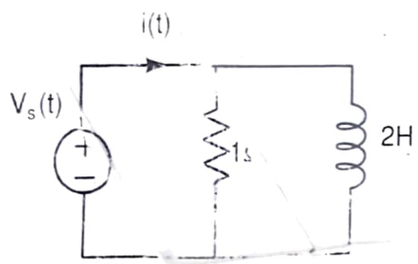
The answers to each question must start on a fresh sheet.
All parts of a question should be answered together
The first sheet must have the name of the instructor.

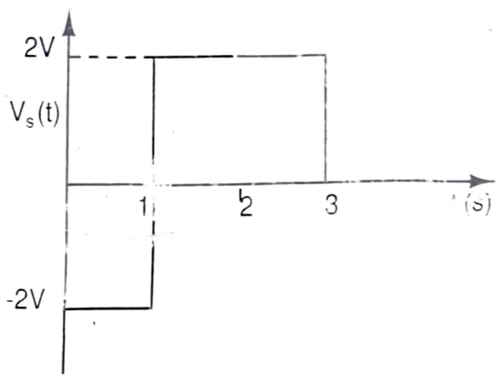
0 (10, 20)

1. Find steady state voltages across the capacitors and currents through the inductors. (2)



2. Assuming zero initial conditions, plot $i(t)$ as a function of time. What is the energy stored in the inductor in steady state? = 0 (2+1)

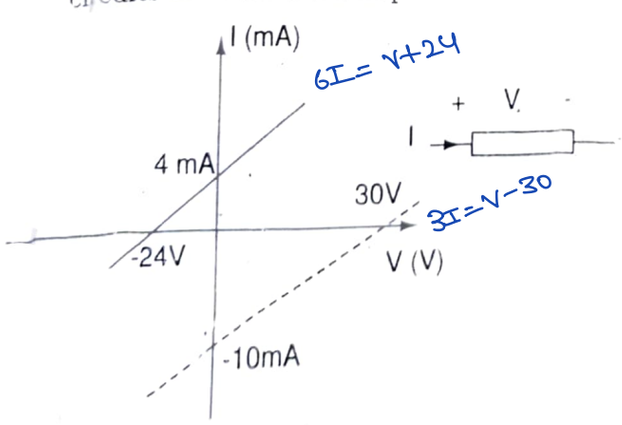


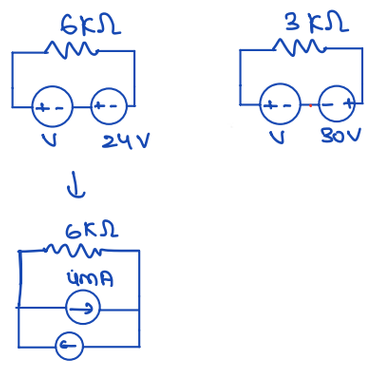


3. The I-V characteristics (with the sign convention as shown in the figure) of two components A and B are given. The x and y intercepts of A (solid line) are -24 V and 4 mA respectively. For B (dashed line) the intercepts are -10 mA and 30 V. (2+2)

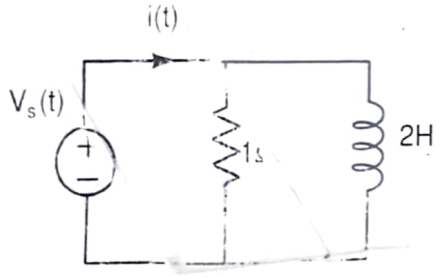
(a) Draw the corresponding circuit diagram for each of the characteristic.

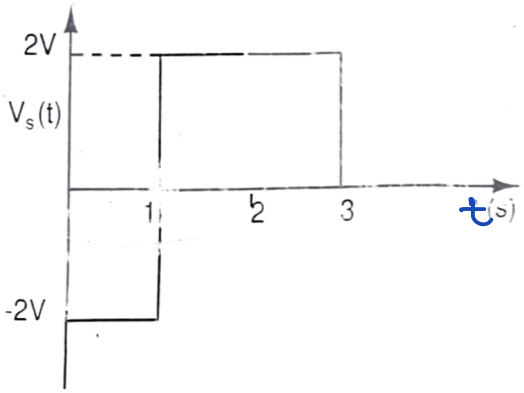
(b) Draw the I-V characteristics, clearly indicating the x and y intercepts, if the two circuits are connected in parallel.

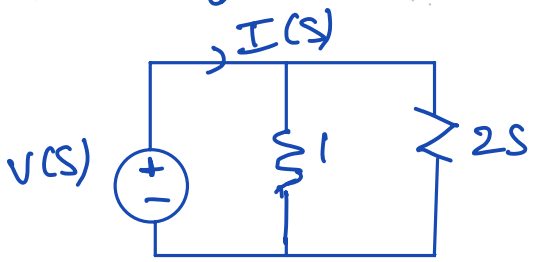




2. Assuming zero initial conditions, plot $i(t)$ as a function of time. What is the energy stored in the inductor in steady state? (2+1)







$$X(s) = \int_0^\infty x(t) e^{-st} dt$$

$$= \int_0^1 (-2) e^{-st} dt + \int_1^3 (2) e^{-st} dt$$

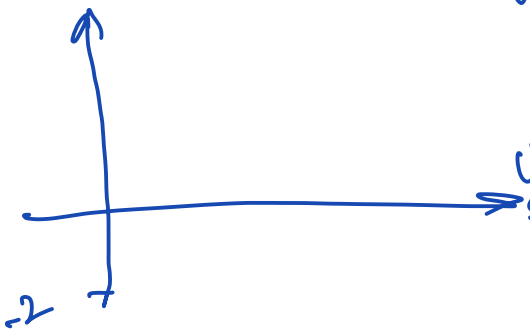
$$\frac{+2}{+s}(e^{-s} - 1) + \frac{2}{-s}(e^{-3s} - e^{-s})$$

$$I(s) \cdot \frac{2s}{2s+1} = \frac{2}{s}\{-e^{-3s} + 2e^{-s} - 1\}$$

$$V(s) = 2/s\{-e^{-3s} + 2e^{-s} - 1\}$$

$$\Rightarrow I(s) = \frac{2s+1}{s^2}\{-e^{-3s} + 2e^{-s} - 1\}$$

$$\Rightarrow$$



$$sI(s)$$

$$\lim_{s\to\infty} I(s) = \frac{2s+1}{s}(-e^{-3s} + 2e^{-s} - 1)$$

$s \to \infty$

$$i(0^+) = (-1)\left(\frac{2+0}{s}\right)$$

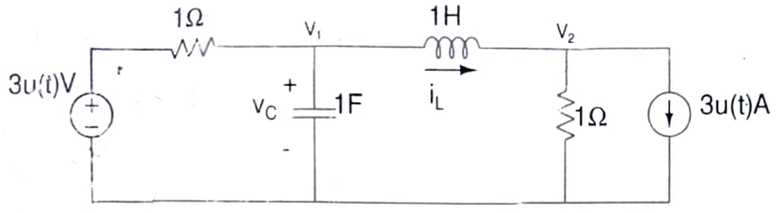
$$i(0^+) = -2\ A$$

$$i(\infty) = \lim_{s\to 0} \frac{2s+1}{s} Ce^{-3s+2e^{-s}-1}$$

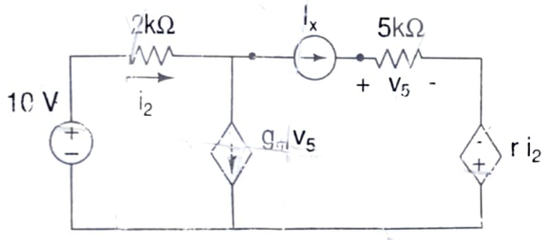
$$\frac{0}{0}$$

$$= \text{DNE}$$

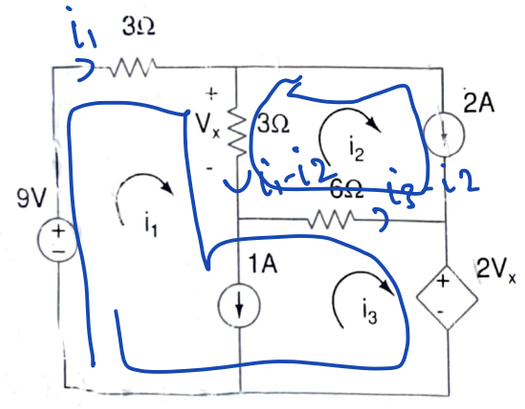
4. In the following circuit, $V_C(0^-) = 2V$ and $i_L(0^-) = iA$. Draw the circuit in the Laplace domain. With the unknowns are $V_1(s)$ and $V_2(s)$, write the nodal equations in Laplace domain and put it in the matrix form. **Do not solve the equations** (2+1)



5. Find the Thevenin equivalent circuit as seen by the independent current source I_x. Also find the value of I_x for which the current source dissipates the maximum power. g_m=1 mS and r=2000 Ω. (2+2)



6. Write the mesh equations for the following circuit, put it in a matrix form and solve for the mesh currents. Clearly indicate supermeshes used in the figure. (2+2)



$9 - 3i_1 - 9(i_1 - i_2) - 6(i_3 - i_2) = 0$

$i_2 = 3$

$i_1 - i_3 = 1$

$9 - 12i_1 + 30 - 6i_3 = 0$

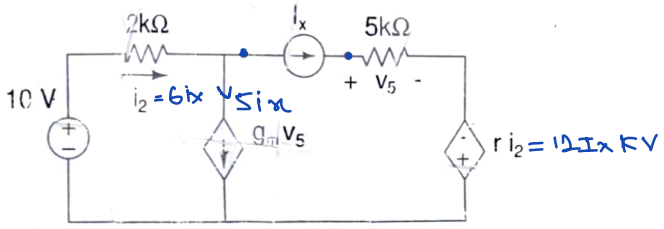
$\Rightarrow 2i_1 + i_3 = 13/2$

$i_1 - i_3 = 1$

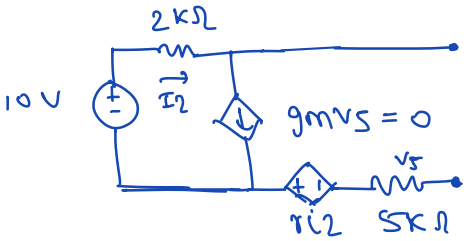
$i_1 = 5/2$

$i_3 = 3/2$

5. Find the Thevenin equivalent circuit as seen by the independent current source I_x. Also find the value of I_x for which the current source dissipates the maximum power. g_m=1 mS and r=2000 Ω. (2+2)

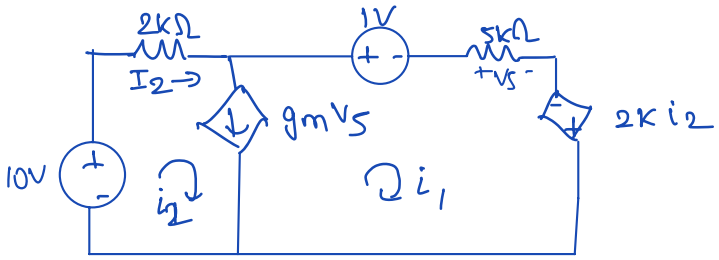


$i_2 - i_x = g_m V_5 = 5 i_x \Rightarrow i_2 - i_x \Rightarrow i_2 = 6 i_x$



$\Rightarrow I_2 = 0 \Rightarrow V_{TH} = 10$

$\Rightarrow V_5 = 0$



$V_5 = 5000\, i_1$

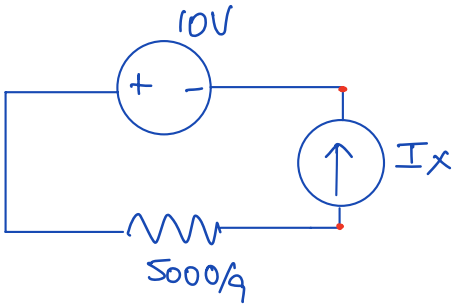
$i_2 - i_1 = V_5/1000$

$-2000 I_2 - 1 - 5000 I_1 + 2000 I_2 = 0$

$\Rightarrow 9 = 5000 I_1$

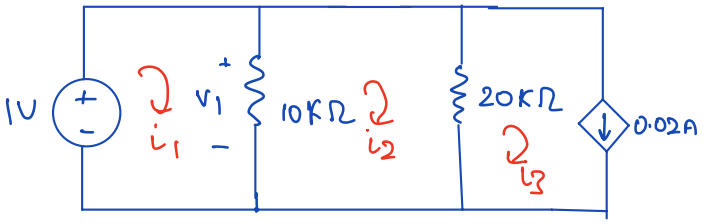
$i_1 = \frac{-1}{5000}$

$R_{TH} = 5K\Omega$



$(10 - 5000 I_x) I_x \Rightarrow 10000 I_x = 10$

$I_x = \frac{1}{1000}$



$$1 = 10^4(i_1 - i_2)$$

$$10^4(i_2 - i_1) = 2\times10^4(i_2 - i_3)$$

$$i_2 - i_1 = 2i_2 - 2i_3$$

$$2i_3 = i_2 + i_1$$

$$i_3 = 2\times10^{-2} \Rightarrow i_1 + i_2 = 4\times10^{-2}$$

$$i_1 - i_2 = 10^{-4}$$

$$2i_1 = 10^{-4}\{1+400\} \Rightarrow i_1 = \frac{401}{2}\times10^{-4}$$

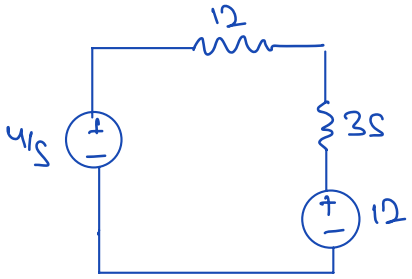
$$R_{TH} = \frac{2\times10^4}{401}$$

$$\frac{2\,di_1(t)}{dt} = v_s(t) = i(t) - i_1(t)$$

$t \to 0$ to 1 $\quad \frac{di_1}{dt} = -1 \rightarrow i_1(t) = -t \Rightarrow i(t) = -t-2$

$t \to 1$ -to- 3 $\quad \frac{di_1}{dt} = 1 \rightarrow i_1(t) = t-2 \Rightarrow i(t) = t$

$t > 3$ $\quad \frac{di_1}{dt} = 0 \rightarrow i_1(t) = 1 \Rightarrow i(t) = 1$



$$4/s - 12I - 3sI - 12 = 0$$

$$\rightarrow (12+3s)I = \frac{4-12s}{s}$$

$$\Rightarrow I = \frac{4}{3}\left\{\frac{1-3s}{s(s+4)}\right\} = \frac{1}{3}\left[\frac{1}{s} - \frac{13}{s+4}\right]$$

$$\Rightarrow i(t) = 1/3\left\{1 - 13e^{-4t}\right\}u(t)$$